Polar Coordinates and Conversions

ANSWER KEY

Converting points

1 Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ $(2, 2\sqrt{3})$

b $\left(2, \frac{3\pi}{4}\right)$ $(-\sqrt{2}, \sqrt{2})$

2 Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$

b $(0, -5)$ $\left(5, \frac{3\pi}{2}\right)$

Converting equations

3 Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

Multiply by r: $r^2 = 6r\cos\theta$, so $x^2 + y^2 = 6x$, i.e. $(x-3)^2 + y^2 = 9$: a circle with center $(3, 0)$ and radius 3.

4 Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ $r = 4$

b $y = x$ $\theta = \frac{\pi}{4}$

5 Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

$r - r\sin\theta = 2$, so $r = y + 2$. Squaring: $x^2 + y^2 = (y+2)^2 = y^2 + 4y + 4$, so $x^2 = 4y + 4$ — a parabola.

Multiple representations

6 Give two other pairs (r, θ) that represent the same point as $\left(3, \frac{\pi}{4}\right)$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.

Adding a full rotation: $\left(3, \frac{\pi}{4} + 2\pi\right) = \left(3, \frac{9\pi}{4}\right)$. Using negative r (add π to the angle): $\left(-3, \frac{\pi}{4} + \pi\right) = \left(-3, \frac{5\pi}{4}\right)$.

7 Plot-reasoning: explain why the polar point $\left(-2, \frac{\pi}{2}\right)$ corresponds to the same location as $\left(2, \frac{3\pi}{2}\right)$.

A negative r means the point lies in the direction opposite to θ. Pointing at angle $\frac{\pi}{2}$ (straight up) with $r = -2$ places the point 2 units in the opposite direction, straight down — the same location as $r = 2$ at angle $\frac{3\pi}{2}$ (also straight down).

Distance between polar points

8 Convert $\left(4, \frac{\pi}{6}\right)$ and $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates, then find the distance between them, to two decimal places.

$\left(4, \frac{\pi}{6}\right) \to (2\sqrt{3}, 2)$; $\left(4, \frac{2\pi}{3}\right) \to (-2, 2\sqrt{3})$. Distance $= \sqrt{(2\sqrt{3}+2)^2 + (2-2\sqrt{3})^2} \approx \sqrt{29.86 + 2.14} = \sqrt{32} \approx 5.66$.

Further conversion practice

9 Convert each polar point to rectangular coordinates (exact values):

a $(6, \pi)$ $(-6, 0)$

b $\left(5, \frac{5\pi}{6}\right)$ $\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$

10 Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.

This is a lemniscate (figure-eight curve). Rewriting $r^2 = 4\sin 2\theta = 8\sin\theta\cos\theta$, and using $r^2 = x^2 + y^2$, $r\cos\theta = x$, $r\sin\theta = y$: multiplying both sides by r^2 gives $(x^2 + y^2)^2 = 8xy$.

Why polar coordinates are not unique

11 Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

Adding any multiple of 2π to θ returns to the same direction, and reversing the sign of r while adding π to θ also gives the same point — so every point has infinitely many valid (r, θ) representations, unlike the one-to-one rectangular system.

12 The pole (origin) is a special case in polar coordinates. Explain why.

The pole corresponds to $r = 0$ for ANY value of θ — every angle gives the same point (the origin) when $r = 0$, so the pole has no single well-defined angle.

Converting more complex equations

13 Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.

$r^2\cos^2\theta - r^2\sin^2\theta = 4 \Rightarrow r^2(\cos^2\theta - \sin^2\theta) = 4 \Rightarrow r^2\cos(2\theta) = 4$.

More point conversions with negative r

14 Convert the polar point $\left(-4, \frac{\pi}{6}\right)$ (negative r) to rectangular coordinates.

$x = r\cos\theta = -4\cos\frac{\pi}{6} = -4 \cdot \frac{\sqrt{3}}{2} = -2\sqrt{3}$. $y = r\sin\theta = -4\sin\frac{\pi}{6} = -4 \cdot \frac{1}{2} = -2$. Point: $(-2\sqrt{3}, -2)$.

15 Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

$r = \sqrt{16+16} = 4\sqrt{2}$. Both coordinates negative places this in QIII, reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

Converting a spiral equation

16 The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.

Since $r = \theta$, the radius grows in direct proportion to the angle swept — each additional full rotation (2π radians) pushes r out by 2π more units, producing evenly-spaced spiral arms rather than a closed curve.

Converting a real-world radar equation

17 A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.

$x = 45\cos\frac{5\pi}{6} = 45\left(-\frac{\sqrt{3}}{2}\right) \approx -38.97$ km. $y = 45\sin\frac{5\pi}{6} = 45\left(\frac{1}{2}\right) = 22.50$ km. The aircraft is about 38.97 km west and 22.50 km north of the station.

18 A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \le \theta < 2\pi$, to two decimal places.

$r = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km. Reference angle $= \tan^{-1}\left(\frac{40}{30}\right) \approx 53.13° \approx 0.93$ rad. Since $x > 0, y < 0$ (QIV): $\theta \approx 2\pi - 0.93 \approx 5.35$ rad.

19 Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r.

$r\cos\theta = 5 \Rightarrow r = \frac{5}{\cos\theta} = 5\sec\theta$. Unlike a circle centered at the pole (constant r), a vertical line's distance from the pole varies with direction θ, so r must depend on θ through the secant function.

20 Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.

$x^2 + y^2 = 36$ (squaring both sides of $r = 6$). There is no θ term because every direction from the pole at distance 6 satisfies the equation equally — the curve is a full circle, independent of angle.